Miao Lin

PH.D. CANDIDATE · DEPARTMENT OF COMPUTER SCIENCE

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Education

Old Dominion University *Norfolk, US*
Ph.D. in Computer Science *08/2023 - Present*

Old Dominion University *Norfolk, US*
Master in Computer Science *08/2023 - 12/2025*

Central South University *Changsha, China*
Master in Medicine *08/2020 - 06/2023*

Research Profile

Efficient & Certified AI

Developing efficient robustness certification for real-world and resource-constrained AI systems, with emphasis on randomized smoothing, adaptive sampling, predictive budget allocation, and systems-aware edge deployment. I evaluate certification methods on heterogeneous platforms, including Jetson Orin Nano, LattePanda Alpha, and Raspberry Pi 4, targeting practical robustness guarantees under constrained computation, memory, and latency.

AI Security: Backdoor & Data Poisoning Defense

Theory-grounded defenses against backdoor and data-poisoning attacks, including certified poisoned-sample detection, robustness under dataset imbalance, and deployment-oriented security evaluation. My work studies how poisoning behavior can be identified through statistically reliable signals and develops defenses that remain effective under realistic data and deployment constraints.

Reliable LLMs & Hallucination Detection

Reliable and efficient verification for large language models, with an emphasis on hallucination detection, adaptive cross-model verification, and instance-level verifier selection.

Teaching & Outreach

08/2026 – Present **Teaching Assistant**, *CS 390: Introduction to Theoretical Computer Science*, Department of Computer Science, ODU. Support instruction and grading; assist students with computability, complexity, and algorithmic foundations.

01/2026 - 12/2026 **Teaching Assistant**, *School of Cybersecurity*, ODU. Designed CTF-style challenges for CyberCup.AI, a unified platform for AI and cybersecurity competitions, focusing on LLM CTF, LLM Backdoor, and AI-Assisted Security scenarios.

09/2025 – 05/2026 **Guest Lecturer**, *CYSE635 (AI Security & Privacy) and CYSE615 (Mobile and Wireless Security)*, ODU. Delivered lectures on adversarial attacks, backdoors, and LLM jailbreaks; led hands-on discussions on attack and defense mechanisms.

07/2024 – 05/2025 **Curriculum Co-designer**, *CYSE635/CS695 (AI Security & Privacy)*, ODU. Co-designed course materials, hands-on labs, and assessments covering adversarial ML, backdoor attacks, differential privacy, and LLM security, with an emphasis on practical attack and defense techniques.

08/2023 – 01/2024, 08/2024 – 01/2025 **Teaching Assistant**, *CS 467/567: Introduction to Reverse Software Engineering*, ODU. Led labs and grading on reverse engineering, API hooking, DLL/process injection, and network analysis; mentored an AI-assisted malware-analysis capstone project.

Publications

Computer Science & AI

M. Lin, Z. Wang, J. Li, F. Yu, L. Li, R. Ning. Instance-Adaptive Cross-Model Verification for Detecting Self-Consistent Hallucinations. *Under review at AAAI 2027.*

M. Lin, F. Yu, R. Ning. Budget-Aware Efficient Certification for Randomized Smoothing on Edge Devices. *Under review at AAAI 2027.*

M. Lin, M. D. Mazumder, F. Yu, L. Li, D. Takabi, R. Ning. Laplace-Bridged Randomized Smoothing for Fast Certified Robustness. *arXiv preprint arXiv:2604.24993, 2026.*

M. Lin, F. Yu, R. Ning, L. Li, Q. Lou, M. Zheng, C. Xin, H. Wu. RPP: a certified poisoned-sample detection framework for backdoor attacks under dataset imbalance. *Transactions on Machine Learning Research (TMLR),* 2026. **J2C Certification.**

M. Lin, J. Zhang, J. Li, F. Yu, L. Li, C. Xin, H. Wu, R. Ning. ACS-Boot: efficient randomized smoothing for robustness certification on resource-constrained edge devices. In *IEEE Conference on Computer Communications* (INFOCOM), 2026.

Z. Chen, J. Li, **M. Lin**, A. Mao, L. Li, R. Ning, C. Xin, H. Wu. TrojanEdge: mutual information–enhanced robust and persistent backdoor attacks for edge and on-device deployments. In *IEEE Conference on Computer Communications* (INFOCOM), 2026.

Previous Biomedical Research

X. Zhang, D. Xiong, L. Deng, *et al.*, **M. Lin**[†]. Lysionotin attenuates bleomycin-induced pulmonary fibrosis by activating AMPK/Nrf2 pathway. *Scientific Reports*. 2025; 15:31306. [†]**Co-corresponding author.**

D. Xiong[#], **M. Lin**[#], D. Lang, *et al.* SLCO4A1-mediated transmembrane transport of lysionotin attenuates acute lung injury by activating the AMPK/Nrf2 signaling pathway. *Phytotherapy Research*. 2025; 39(10):4404–4427.

L. Deng, W. Xie, **M. Lin**, *et al.* Taraxerone inhibits M1 polarization and alleviates sepsis-induced acute lung injury by activating SIRT1. *Chinese Medicine,* 2024, 19:159.

W. Xie, L. Deng, **M. Lin**, *et al.* Sirtuin 1 mediates the protective effects of echinacoside against sepsis-induced acute lung injury by regulating the NOX4–Nrf2 axis. *Antioxidants*. 2023; 12(11):1925.

M. Lin, W. Xie, D. Xiong, *et al.* Cyasterone ameliorates sepsis-related acute lung injury via AKT(Ser473)/GSK3β(Ser9)/Nrf2 pathway. *Chinese Medicine*. 2023; 18:136.

Academic Service

- **Reviewer**: ICNC, Journal of Information Security and Applications. *2026*
- **Entrepreneur Lead**, NSF Regional I-Corps Program. *05/2025 – 10/2025*
  Conducted 40+ customer-discovery interviews on privacy-technology needs and refined problem-solution fit, value proposition, and go-to-market strategy for a cybersecurity startup concept.

Awards & Honors

- **Student Travel Awards, IEEE INFOCOM** *2026*
- **2024-2025 Dominion Scholar, ODU** *2024-2025*
- **Second Prize Scholarship, CSU** *2020-2022*
- **Provincial Excellent Undergraduate** *2020*
- **National Scholarship** *2019*